

CryptoPilot

Automated BTC Trading Platform — Business Solution Document

Automated execution of the validated Trend Rider v5.2 strategy on Binance USDT-M Futures, with a web dashboard, SMS alerts, pluggable strategies and an AI market-news assistant.

Field	Value
Document	Business Solution Document (BSD)
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Prepared for	Project owner (Binance account holder)
Stack	Python 3.11+ · FastAPI · PostgreSQL · React · Docker
Trading venue	Binance USDT-M Futures (Demo/Testnet first, then Live)

IMPORTANT — RISK NOTICE: This document describes software that trades real money automatically. Backtested performance (2023–2026) does not guarantee future results. Leveraged futures trading can lose more than the amount risked per trade. The system described here enforces risk controls, but no automated system removes market risk.

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1. Executive summary

CryptoPilot is a self-hosted automated trading platform that executes the owner's validated 4-hour BTCUSDT trend-following strategy (Trend Rider v5.2) on Binance, unattended, 24/7. The owner controls the bot from a secure web dashboard: start/stop it at any time, switch between Binance Demo (testnet) and Live accounts, watch open positions and live profit, review every historical trade with win/loss detail, and see how much of the current month's profit is available to withdraw. Every significant event — trade opened, trade closed, bot started, bot stopped, errors — is pushed to the owner's phone by SMS through notify.lk.

The platform is deliberately built in two decoupled halves. The **trading core** (strategy engine + execution engine) is strategy-agnostic: strategies are plugins behind a fixed interface, so Trend Rider v5.2 can be replaced or A/B-swapped without touching execution, risk, or dashboard code. The **intelligence layer** is a separate AI news agent that collects daily crypto/macro news, summarises it with the Claude API, and presents a morning briefing on the dashboard. The news agent informs the human; it never places trades.

Rollout is safety-first: the system runs against the official Binance Futures testnet with fake money until its live decisions match the validated backtest engine, and only then is switched to the live account — starting at conservative position sizing (1–5% risk per trade).

Verdict on feasibility: **high**. Every required capability is a documented, officially supported Binance API feature; the strategy acts only on closed 4-hour candles, which makes timing, rate limits and infrastructure demands modest.

2. Business goals and success criteria

#	Goal	Success criterion
G1	Fully automated trading of the validated strategy	Bot replicates backtest decisions on identical candles with 100% rule fidelity
G2	Owner control and visibility	Start/stop from dashboard; live position, balance and P&L visible at all times
G3	Safe path to production	≥ 2–4 weeks on testnet with zero unexplained deviations before live keys
G4	Immediate awareness	SMS delivered for every trade open/close and bot start/stop within ~1 minute
G5	Strategy agility	A new strategy can be added as a plugin without modifying core services
G6	Informed trading context	Daily AI news briefing available on the dashboard by a fixed local time
G7	Withdrawal discipline	Dashboard computes the month's withdrawable profit; withdrawals stay manual

3. Scope

In scope (v1)

- Single exchange (Binance USDT-M Futures), single symbol (BTCUSDT), long-only strategy.
- Environment switching between Binance Demo/Testnet and Live via dashboard settings.
- Trend Rider v5.2 as the first strategy plugin (regime filter, ATR stop, TP1 40%, breakeven, 4×ATR trail, regime-death exit, 4% monthly circuit breaker).
- Risk-based position sizing (configurable % of equity risked per trade, leverage cap).

- Web dashboard with authentication: bot control, live position, equity curve, trade history, monthly P&L and withdrawable amount, news briefing.
- SMS notifications via notify.lk; full event log in the database.
- AI news agent with daily scheduled collection and Claude-generated summaries.
- Docker-based deployment on a small always-on VPS.

Out of scope (v1)

- Automatic withdrawals or any transfer of funds (deliberately excluded; withdrawal permission is never granted to the API key).
- Short selling, multi-coin portfolios, grid/HFT strategies.
- Mobile app (the dashboard is responsive; SMS covers push needs).
- News-driven automated trading decisions (news is informational only).

4. System architecture

CryptoPilot is a modular monolith: one deployable backend with strictly separated internal services, plus a React single-page dashboard. Modules communicate through defined interfaces and the PostgreSQL database — never by reaching into each other's internals — so any module (strategy, notifier, news agent) can be replaced in isolation.

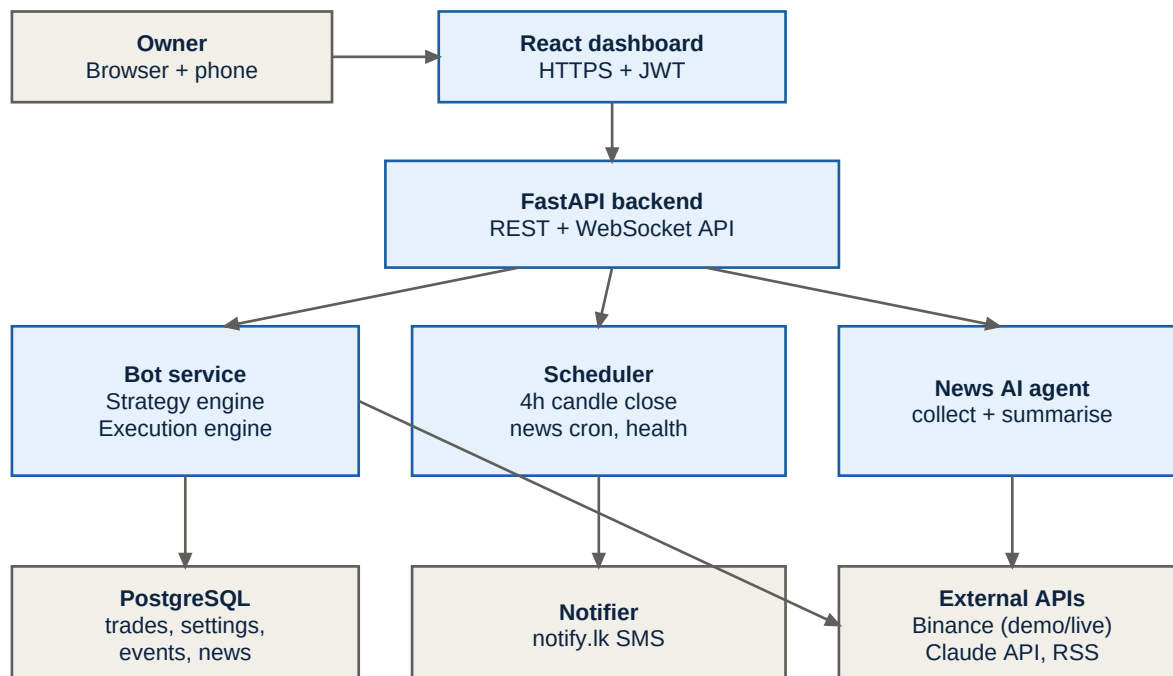


Figure 1 — Component architecture. Solid arrows = primary data/control flow.

Component responsibilities

Component	Responsibility
React dashboard	UI: login, bot control, settings (environment, risk %), live position, equity & trade history, withdrawal figure, news briefing.
FastAPI backend	Auth (JWT), REST API for the dashboard, WebSocket push for live updates, orchestrates bot lifecycle commands.
Bot service	The 24/7 trading loop. Subscribes to Binance kline stream, runs the active strategy plugin on each closed 4h candle, hands orders to the execution engine, reconciles state with Binance on every start.
Strategy engine	Loads the configured strategy plugin; feeds it candles; receives abstract intents (enter, move stop, take profit, exit, halt).
Execution engine	Translates intents into Binance orders (MARKET, STOP_MARKET, TAKE_PROFIT_MARKET), enforces lot-size/min-notional filters, leverage cap, and the 'stop always resting on exchange' rule.
Scheduler	Candle-close ticks, daily news run, watchdog health checks, equity snapshots.
Notifier	Formats and sends SMS via notify.lk; retries; logs every notification.
News AI agent	Fetches configured feeds, deduplicates, summarises via Claude API, stores the daily briefing.

Component	Responsibility
PostgreSQL	Single source of truth: settings, strategies, runs, trades, orders, equity snapshots, events, news.

5. Functional requirements

ID	Requirement	Priority
FR-01	Settings page selects the active environment: DEMO (testnet) or LIVE. Each environment stores its own API key pair. Switching environments requires the bot to be stopped and shows a confirmation dialog.	Must
FR-02	Dashboard Start/Stop button controls the bot. Start = bot connects, reconciles account state, begins 24/7 loop. Stop = bot cancels nothing by default but stops evaluating; an optional 'Stop & close position' action flattens safely. State survives server restarts (auto-resume if it was running).	Must
FR-03	Bot trades Trend Rider v5.2 rules exactly as validated (Appendix A), acting only on closed 4h candles.	Must
FR-04	Position sizing: risk% of current equity per trade (default 2% on live, configurable), leverage capped (default 3x, hard max configurable), quantity rounded to Binance lot size, minimum-notional respected.	Must
FR-05	Portfolio data (balance, positions, income history) is read from the Binance account API and displayed on the dashboard; the account is the source of truth, not the bot's memory.	Must
FR-06	SMS via notify.lk for: trade opened, TP1 filled, trade closed (with P&L), bot started, bot stopped, circuit-breaker halt, and any execution error. Message templates are configurable.	Must
FR-07	Trade history view: every trade with entry/exit time and price, size, fees, realized P&L, R-multiple, exit reason (stop / TP1+trail / regime death / breaker), win/loss flag; filter and CSV export.	Must
FR-08	Monthly view: profit per calendar month and 'withdrawable this month' figure computed by the configured rule (default: 10% of the month's realized profit, only when positive). Withdrawals remain manual.	Must
FR-09	Strategy plugin framework: strategies are discoverable classes implementing a fixed interface; active strategy chosen in settings; per-strategy parameters stored in DB.	Must
FR-10	AI news agent: on a daily schedule (and on demand) fetch news sources, summarise into a briefing (major crypto/macro events, sentiment, calendar items like FOMC/CPI dates), display on dashboard with source links.	Must
FR-11	Event log screen: every bot decision and system event, timestamped, searchable.	Should
FR-12	Equity curve chart including monthly deposits/withdrawals annotations.	Should
FR-13	Kill-switch: a single 'flatten everything and stop' button, protected by confirmation.	Must

6. Non-functional requirements

Area	Requirement
Reliability	Bot must survive restarts without losing track of positions: on every start it reconciles against Binance (open position? resting orders?) and resumes. Protective stop orders always rest on the exchange so the account is protected even if the server dies.
Availability	Target 99.5%+ for the bot loop. A watchdog restarts the bot process; missed-candle detection backfills from the REST kline endpoint.
Latency	Actions occur within seconds of a 4h candle close — ample for this strategy; no HFT requirements.
Security	See §13. Keys never leave the server; no withdrawal permission; JWT dashboard auth; HTTPS only.

Area	Requirement
Auditability	Every decision, order, fill, SMS and error is persisted with timestamps; trades reconcile to Binance income history.
Maintainability	Strategy, notifier and news modules replaceable behind interfaces; typed Python (mypy), tests for strategy math against the validated backtest engine.
Portability	Docker Compose; runs on any \$5–10/month VPS (1 vCPU, 1–2 GB RAM).

7. Strategy plugin framework (loose coupling)

The strategy never talks to Binance and never sizes positions. It receives candles and current trade state, and returns abstract **intents**. The execution engine owns sizing, exchange filters and order mechanics. This is what makes strategies swappable: a new idea is a new class, registered by name, selectable in settings — zero changes to execution, risk, dashboard or notifications.

```
class Strategy(Protocol):                                # strategies/base.py
    name: str
    params: dict                                         # editable in dashboard settings

    def warmup_bars(self) -> int: ...                    # e.g. 200 for SMA200

    def on_candle(self, candles: DataFrame,
                   state: TradeState) -> list[Intent]: ...

# Intents (the ONLY vocabulary strategies may speak)
EnterLong(stop_distance: float, tp_levels: list[TPLevel])
MoveStop(price: float)                                # ratchet-only enforced by engine
TakePartial(level_id: str)                             # informational; TP orders rest on exchange
ExitAll(reason: str)                                   # regime death, breaker, manual
Halt(until: datetime, reason: str)                     # monthly circuit breaker

# Registration
STRATEGIES = {"trend_rider_v52": TrendRiderV52} # add new entries here
```

Guarantees enforced by the engine regardless of strategy: stops only move up (ratchet), a protective stop must exist whenever a position exists, leverage cap and lot filters are applied after sizing, and the monthly circuit breaker is engine-level so a faulty strategy cannot disable it.

8. Binance integration design

Concern	Design
Environments	DEMO: https://demo-fapi.binance.com (REST), demo WebSocket stream. LIVE: https://fapi.binance.com . Selected via settings (FR-01); separate key pairs; the active environment is shown permanently in the dashboard header.
Market data	WebSocket kline stream (4h) with closed-candle flag; REST klines for warmup/backfill and gap repair after disconnects.
Order placement	Entry: MARKET. Protection: STOP_MARKET (closePosition=true) resting on exchange. TP1: TAKE_PROFIT_MARKET reduceOnly for 40% of the position. Trailing: bot cancels/replaces the STOP_MARKET once per closed candle.
Fills & account	User Data Stream WebSocket for real-time fill events; REST account, positionRisk and income endpoints for balances, portfolio and P&L shown on the dashboard (FR-05).
Auth	HMAC-SHA256 signed requests, timestamp + recvWindow; server clock synced via NTP.
Rate limits	Bot usage is a handful of calls per 4 hours — orders of magnitude below limits; 429/ban handling implemented anyway with exponential backoff.
Reconciliation	On start and every 15 minutes: fetch position + open orders; if DB state and exchange state disagree, exchange wins, discrepancy is logged and SMS-alerted.

9. Database design (PostgreSQL)

PostgreSQL 16. All timestamps UTC (timestampz). Numeric(20,8) for prices/quantities — never floats for money. Key tables:

Table	Purpose / key columns
users	Dashboard login. id, email, password_hash (argon2), role, created_at.
app_settings	Singleton config. active_environment (DEMO LIVE), active_strategy, risk_pct, leverage_cap, withdraw_rule, sms_enabled, news_sources[].
api_credentials	Per-environment key pairs. environment, api_key, api_secret_encrypted (AES-GCM via app master key from env), created_at.
strategies	Registered plugins. name, class_path, params_json, enabled.
bot_runs	One row per start→stop. started_at, stopped_at, environment, strategy, stop_reason (user error kill), started_by.
candles	Cached klines. symbol, interval, open_time PK, o/h/l/c/v.
trades	One row per round-trip. opened_at, closed_at, side, entry_px, exit_px, qty, entry_fee, exit_fee, realized_pnl, r_multiple, exit_reason, is_win, strategy, environment, bot_run_id FK.
orders	Every exchange order. binance_order_id, trade_id FK, type, status, price, stop_price, qty, reduce_only, placed_at, filled_at, raw_json.
equity_snapshots	Periodic account equity for the curve. ts, balance, unrealized_pnl, environment.
monthly_ledger	Per calendar month: realized_pnl, fees, withdrawable_amount, withdrawn_amount (manual entry), breaker_triggered.
events	Audit log. ts, level, category (trade bot system news sms), message, data_json.
notifications	SMS log. ts, phone, template, body, notify_lk_status, retries.
news_items	Raw fetched items. source, url UNIQUE, title, published_at, raw_text.
news_briefings	Daily AI output. date, summary_md, sentiment, model, input_item_ids[], created_at.

Migrations via Alembic. Nightly pg_dump to off-server storage; the database plus Binance's own records fully reconstruct system history.

10. Notification service — notify.lk SMS

notify.lk exposes a simple authenticated HTTPS API (user id + API key + sender id, message and recipient number). The Notifier is an internal queue consumer: services emit typed events; the notifier renders a template, calls notify.lk, retries up to 3 times with backoff, and records delivery status in the notifications table. SMS failure never blocks trading — it is logged and shown as a dashboard warning banner.

Event	Example SMS
Trade opened	CryptoPilot [LIVE]: BUY 0.004 BTC @ 68,894. Stop 66,586, TP1 71,202. Risk 2%.
TP1 filled	CryptoPilot: TP1 hit @ 71,202. Sold 40%. Stop moved to entry — trade now risk-free.
Trade closed	CryptoPilot: Trade closed @ 94,972. P&L +\$205.10 (+20.5%). Reason: trailing stop.
Bot started	CryptoPilot: Bot STARTED on DEMO. Strategy: trend_rider_v52, risk 2%.
Bot stopped	CryptoPilot: Bot STOPPED by user. No open position.

Event	Example SMS
Breaker	CryptoPilot: Monthly loss cap hit (-4%). All closed. Halted until 1st.
Error	CryptoPilot ALERT: order rejected (code -2019 margin). Bot paused, check dashboard.

11. AI news assistant

A fully separate module — it shares only the database and scheduler with the rest of the system, and has no access to trading or keys. Pipeline (daily at a configured local time, plus on-demand refresh):

- **Collect** — fetch configured sources: crypto RSS feeds (e.g. CoinDesk, CoinTelegraph, Bitcoin Magazine), macro headlines, plus the static economic calendar (FOMC decision dates, CPI release dates).
- **Deduplicate & store** — URL-unique insert into news_items.
- **Summarise** — send the day's items to the Claude API (claude-haiku-4-5 for cost efficiency; claude-sonnet-5 configurable for deeper briefings) with a fixed prompt: 5–8 bullet major events, market-relevant only, neutral tone, flag anything affecting BTC volatility (regulation, ETF flows, Fed).
- **Publish** — store briefing; dashboard shows it with source links and the upcoming FOMC/CPI dates panel.

Design rule: the agent is read-only context for the human. Backtests showed event-based trade filtering loses money, so news never gates or triggers orders.

12. Dashboard specification

Screen	Contents
Login	Email + password (argon2), JWT session, optional TOTP 2FA.
Overview	Environment badge (DEMO/LIVE), bot status + Start/Stop + kill-switch, account balance, open position card (entry, size, current stop, TP1, unrealized P&L, R progress), this-month profit, withdrawable amount, equity curve, today's news briefing.
Trades	Filterable table of all trades (FR-07) with per-trade detail drawer showing its orders and fills; CSV export.
Monthly	Calendar-month table: trades, realized P&L, fees, breaker status, withdrawable amount, mark-withdrawn action (records manual withdrawal).
News	Daily briefings archive, upcoming FOMC/CPI panel, source links.
Events	Audit log with level/category filters.
Settings	Environment switch (bot must be stopped), API key entry (write-only fields, never displayed back), strategy selection + parameters, risk %, leverage cap, SMS templates/toggles, news sources, withdraw rule.

Live updates via WebSocket (position, P&L, bot state) so the overview is real-time without refreshing.

13. Security design

- **Binance API keys:** created by the owner only after the futures account is active; permissions = Futures trading only; **withdrawals never enabled**; key restricted to the VPS static IP. Keys entered via write-only settings fields, encrypted at rest (AES-GCM with a master key supplied by environment variable), never logged, never returned by any API.
- **Transport:** HTTPS only (Caddy/Let's Encrypt), HSTS; dashboard behind JWT with short-lived tokens; optional IP allowlist for the dashboard itself.
- **Server:** minimal VPS, SSH keys only, UFW firewall (443 + SSH), automatic security updates, fail2ban.
- **Application:** role-based (single owner role in v1), rate-limited login, audit of every settings change, secrets excluded from backups' plaintext (encrypted fields stay encrypted in dumps).
- **Blast-radius design:** even full server compromise cannot withdraw funds (no withdrawal permission + IP lock); worst case is bad trades, bounded by the leverage cap and circuit breaker.

14. Testing and rollout plan

Phase	What happens	Exit criteria
P1 — Unit & parity	Strategy math tested against the validated Python backtest on 3 years of stored candles.	Bit-identical decisions on all 6,800+ candles
P2 — Testnet live	Bot runs 24/7 on Binance demo with real market data and fake money; SMS and dashboard fully active.	2–4 weeks; every trade explained; zero unhandled errors; restart/reconcile drills passed
P3 — Chaos drills	Deliberate kill of bot mid-trade, network cuts, Binance maintenance windows, clock skew.	Position always protected by resting stop; clean recovery every time
P4 — Live pilot	Live account, minimum viable capital, risk 1–2%.	4+ weeks matching expectations; owner comfortable

Phase	What happens	Exit criteria
P5 — Normal operation	Risk raised gradually toward target (recommended ceiling 5%).	Ongoing monthly review vs backtest expectations

15. Deployment and operations

Docker Compose stack on one VPS: **app** (FastAPI + bot + scheduler + news agent), **db** (PostgreSQL 16 + nightly dump), **caddy** (TLS reverse proxy). The dashboard is a static React build served by Caddy. Ops runbook covers: restart procedure, key rotation, environment switch, restoring from backup, and the manual kill-switch. Monitoring: healthcheck endpoint + an external uptime ping (e.g. UptimeRobot) that SMS-alerts if the app goes dark — covering the failure mode the app cannot report itself.

16. Project plan and estimates

Milestone	Deliverables	Est. effort
M1 — Foundations	Repo, Docker, DB schema + migrations, auth, settings, environment switching	1–1.5 wk
M2 — Trading core	Strategy plugin framework, Trend Rider v5.2 plugin, execution engine, reconciliation, candle pipeline	2 wk
M3 — Control & alerts	Start/stop lifecycle, watchdog, notify.lk notifier + templates, event log	1 wk
M4 — Dashboard	Overview, trades, monthly, settings screens; WebSocket live updates; equity curve	1.5–2 wk
M5 — News agent	Collectors, Claude summarisation, briefing UI, calendar panel	1 wk
M6 — Hardening	Parity tests (P1), chaos drills (P3), security pass, docs/runbook	1 wk
Testnet soak (P2)	Calendar time in parallel with M4–M6	2–4 wk elapsed

Total build effort ≈ 7–8.5 working weeks; go-live after testnet soak.

17. Operating costs (monthly estimate)

Item	Estimate	Notes
VPS (1–2 GB)	\$6–12	Hetzner / DigitalOcean / Vultr
Domain + TLS	~\$1	TLS free via Let's Encrypt
notify.lk SMS	\$1–5	~30–60 SMS/month at typical LKR rates; prepaid credits
Claude API (news)	\$1–5	1 daily briefing on claude-haiku-4-5 is pennies; higher if claude-sonnet-5 selected
Backups (object storage)	\$1–2	Nightly encrypted dumps
Binance fees	Variable	0.02%/0.05% maker/taker on traded notional — already included in all strategy backtests
Total fixed	≈ \$10–25 / month	

18. Risks and honest limitations

Risk	Reality / mitigation
Market risk	The strategy's validated edge includes losing months and multi-month flat periods (bear markets ≈ capital preservation, not profit). Backtest: worst month −5.6% at full sizing; out-of-sample bear year −4.1%. No mitigation removes this — sizing controls it.
Over-leverage / liquidation	At high risk settings ($\geq 10\text{--}20\%$ /trade) effective leverage reaches 5–13x where a fast wick can liquidate before a stop fills — this is NOT modeled in backtests. Mitigation: default risk 2%, recommended ceiling 5%, hard leverage cap in the execution engine.
Execution gaps vs backtest	Slippage on market orders and stop wick-throughs mean live results will differ slightly from backtest fills. Mitigation: conservative fill assumptions were already used; monthly live-vs-expected review.
Infrastructure failure	Server/network death mid-trade. Mitigation: exchange-resting stops (account protected while bot is dark), watchdog auto-restart, reconcile-on-start, external uptime alerting, SMS on recovery.
Exchange risk	Binance outages, API changes, regional access issues. Mitigation: resting stops, changelog monitoring, versioned client, kill-switch.
Key compromise	Mitigated to trade-only blast radius (no withdrawal permission, IP lock, encryption at rest); rotation runbook.
Automation trust	The most common real-world failure is the owner overriding the bot mid-drawdown. Mitigation: dashboard shows expected-vs-actual context; all manual interventions logged and SMS-confirmed.

19. Future enhancements (post-v1)

- Multi-strategy concurrent allocation (capital split across plugins).
- Additional symbols/exchanges behind the same execution interface.
- Telegram/WhatsApp channel in the notifier alongside SMS.
- Walk-forward re-validation job that re-runs the backtest monthly on fresh data and flags edge degradation on the dashboard.
- Paper-trading mode replaying live candles against a candidate strategy in parallel with the active one (shadow testing).

Appendix A — Trend Rider v5.2 strategy summary

Validated on real Binance BTCUSDT 4h data, June 2023 → July 2026, fees included, conservative fills. Full research trail: BACKTEST_REPORT.md, IMPROVEMENT_REPORT.md, TRADING_GUIDE.md in the project repository.

Rule	Definition
Regime filter	Long only while close > SMA200 AND EMA50 > EMA200 (4h).
Entries	Regime turns on; or pullback below EMA20 then close back above, while in regime. Acted on next candle open.
Initial stop	Entry − 2.5 × ATR(14).
TP1	At +1R sell 40%; move stop to breakeven.

Rule	Definition
Trail	Runner trails at highest-high – 4 × ATR (ratchet up only).
Hard exits	Regime flips off → close all at next open. Monthly circuit breaker: equity –4% in a calendar month → close all, halt until the 1st.

Metric (3 years)	v5.2	Buy & hold
Total return	+137.3%	+134.6%
Sharpe	1.45	~0.8
Max drawdown	–13.2%	–53%
Profit factor / win rate	1.84 / 48% (100 trades)	—
Worst month	–5.6%	—
Out-of-sample bear year	–4.1%	–45%

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